

Isogeometric Analysis — Practice 4

Question 1 Recover knot spans from Bézier extraction row operators

Write a program to convert from list of Bézier extraction row operators to knot spans that represent a basis function. The polynomial order can be determined from the size of a Bézier extraction operator.

```
def recoverKnotSpans(CRs):  
    assert(len(CRs)>0)  
    p = len(CRs[0]) - 1  
    knot_spans = numpy.empty((p+1, p+1), dtype="f8")  
    #...  
    return knot_spans
```

Question 2 Connect a cable to a membrane

If you have difficulty doing this, you can read the paper [1] and create the same mesh explained in the appendix. As a mesh representation, you can use “Version 4” format from “File_Format.pdf.”

References

- [1] T. Terahara, K. Takizawa, and T.E. Tezduyar, “T-splines computational membrane–cable structural mechanics with continuity and smoothness: I. Method and implementation”, *Computational Mechanics*, **71** (2023) 657–675, doi:[10.1007/s00466-022-02256-w](https://doi.org/10.1007/s00466-022-02256-w).